

Arunabha Majumder

Designer and Researcher | Multidisciplinary Research | Quantitative Methods | Data Visualization

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PROFESSIONAL SUMMARY

Designer and researcher with a Master's degree (M.Tech) in Mechatronics and a PhD in progress at Aalborg University, with more than five years of research experience in multidisciplinary projects that combine design, human-subject research and data analysis. Experienced in contributing to the implementation of research plans end to end: study design, ethics protocols, quantitative data collection and statistical analysis, through to peer-reviewed publications (Best Research Paper Award, 2026). Secured an INR 100,000 innovation grant. Trained visual artist (Diploma in Fine Arts) who creates illustrations, technical visuals and data visualizations. Experienced in international research environments in Denmark and India; fluent in English (C1).

CORE COMPETENCIES

- **Research:** Contributing to the implementation of research plans, multidisciplinary research, experimental study design, human-subject studies, ethics protocols
- **Research Methods:** Quantitative research methods, statistical power analysis, statistical analysis, data collection (EMG, motion capture), end-to-end data pipelines
- **Data Analysis & Visualization:** Python, MATLAB, visualization of experimental results; Adobe Illustrator and Photoshop for figures and visuals
- **Design:** Concept development, iterative design, prototyping, system design, testing with users
- **Communication & Content:** Scientific writing and peer-reviewed publications, grant funding (TATA Technologies Innovation Grant), illustration and graphic design, English (C1)
- **Collaboration:** Collaboration in multidisciplinary and international teams (engineering, controls, biomechanics)

RESEARCH EXPERIENCE

PhD Researcher – Design and Research of Assistive Robotic Systems (VIXO Project) | 2023 – Present

Aalborg University, Department of Materials and Production, Aalborg, Denmark

- Contribute to implementing the research plan of the VIXO research project, taking research questions from concept and design through experimental validation to publication.
- Designed and conducted a 12-participant human study across 3 assistance modes and 2 load conditions, covering statistical power justification, EMG and motion-capture instrumentation, ethics protocol and the end-to-end data pipeline; the dataset supports two journal submissions.
- Conduct multidisciplinary research that combines mechanical design, electronics, control and biomechanics, working with co-authors from several disciplines.
- Designed a hybrid-actuated shoulder exoskeleton that cut energy consumption by more than 25%; the work won the Best Research Paper Award at IFToMM ISRM 2026.
- Reduced user muscle effort by more than 15% with an Assist-as-Needed controller, validated with human participants.
- Developed analytical models and validated them against experimental data to below 10% RMS error, and proposed a sensorless force-estimation method that reduced system cost, mass and complexity.

M.Tech Researcher – Actuator Systems (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India

- Carried out a quantitative experimental research project: built test setups, collected and analysed data, and developed an empirical actuator model.

- Developed a neural-network-based controller that reduced position-tracking error to 0.3–0.78%; published at IEEE CONECCT 2022.

Bachelor's Capstone Project – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019

Siddaganga Institute of Technology, Tumakuru, India

- Secured an INR 100,000 TATA Technologies Innovation Grant and won the Best Major Project Award for a rehabilitation device, leading design, fabrication and user testing end to end.

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present

Aalborg University, Department of Materials and Production, Aalborg, Denmark

Master's Degree (M.Tech), Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

Bachelor's Degree (B.Tech), Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

Diploma in Fine Arts (Painting)

Bangiya Sangeet Parishad, India

PUBLICATIONS

- Majumder, A., Wagner, J.W., Zhu, Y., Oliveira, A.S., and Bai, S. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." *Robotics and Mechatronics: ISRM 2026, MMS*, vol. 213, Springer. (Best Research Paper Award)
- Majumder, A., Sarkar, D., Chakraborty, S., Singh, A., Roy, S.S., and Arora, A. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." *IEEE CONECCT*, 2022.

GRANTS & AWARDS

- Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics (2026)
- TATA Technologies Innovation Grant – INR 100,000 (2019)
- Best Major Project Award, Department of Mechanical Engineering (2019)
- Certified SolidWorks Associate (CSWA)

TEAM PROJECTS

- **Efficycle (2018):** Worked in a multidisciplinary team to design a human-electric hybrid vehicle for the Efficycle competition.
- **Go-Kart (2017):** Led end-to-end design and fabrication of a custom Go-Kart.

TOOLS

- **Data & Analysis:** Python, MATLAB/Simulink, LabVIEW (basic), LaTeX
- **Visual Content:** Adobe Illustrator, Adobe Photoshop, Procreate (basic)
- **Research Instrumentation:** Qualisys Motion Capture, EMG, force/torque sensors
- **Design & Prototyping:** SolidWorks, Fusion 360, 3D printing

LANGUAGES

English (C1, fluent) | Bengali (Native) | Hindi (Conversational)

REFERENCES

Available upon request.